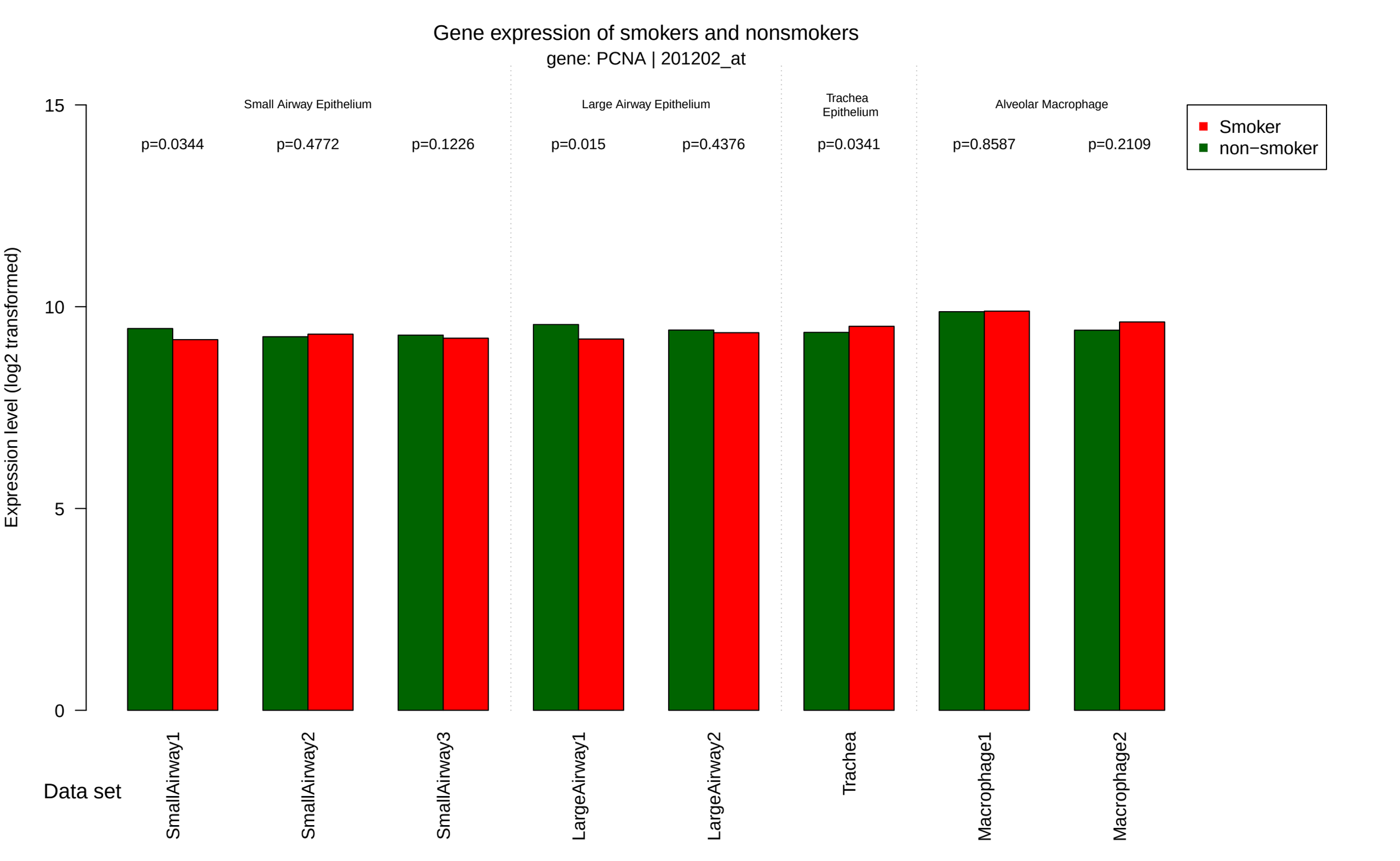
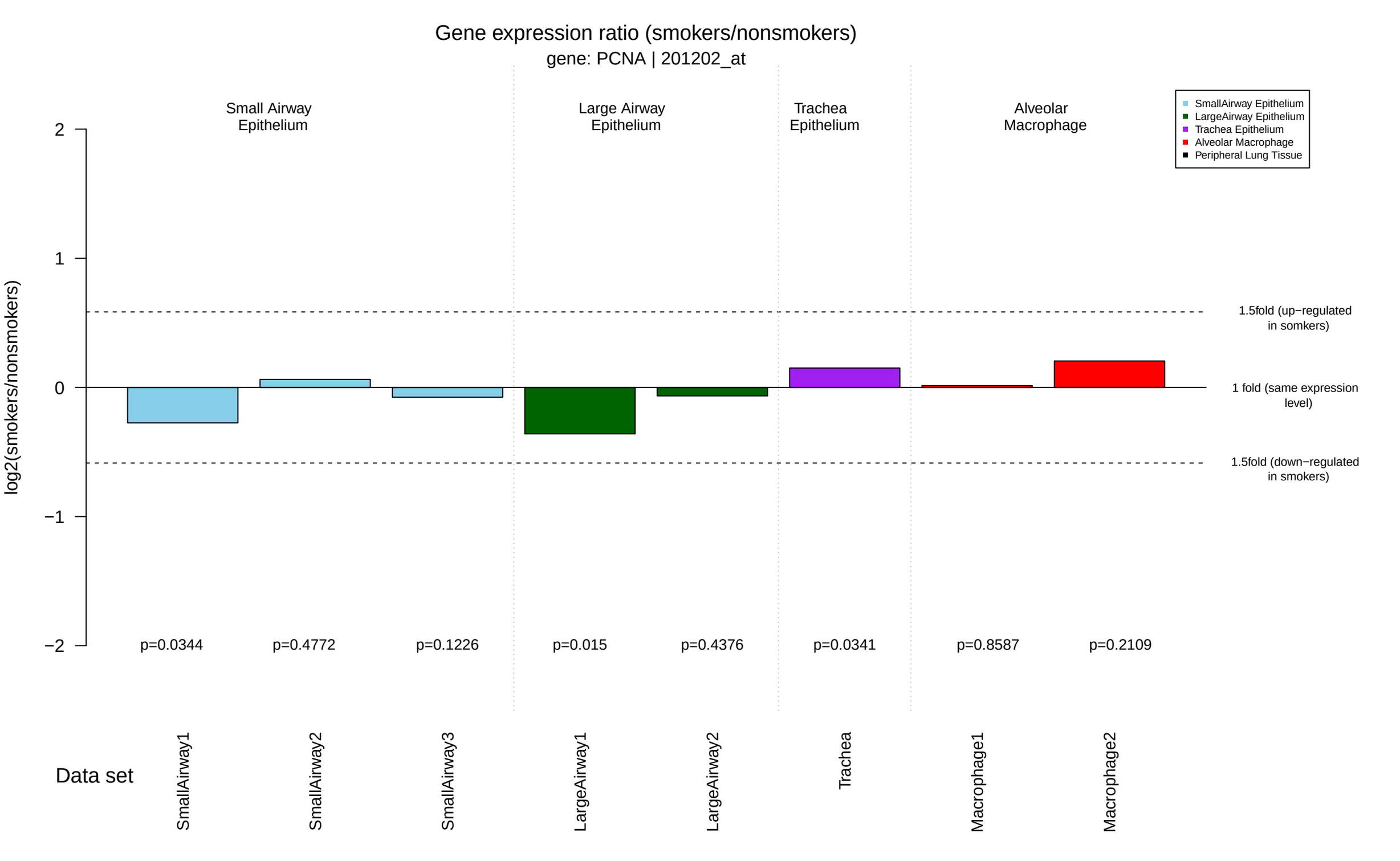
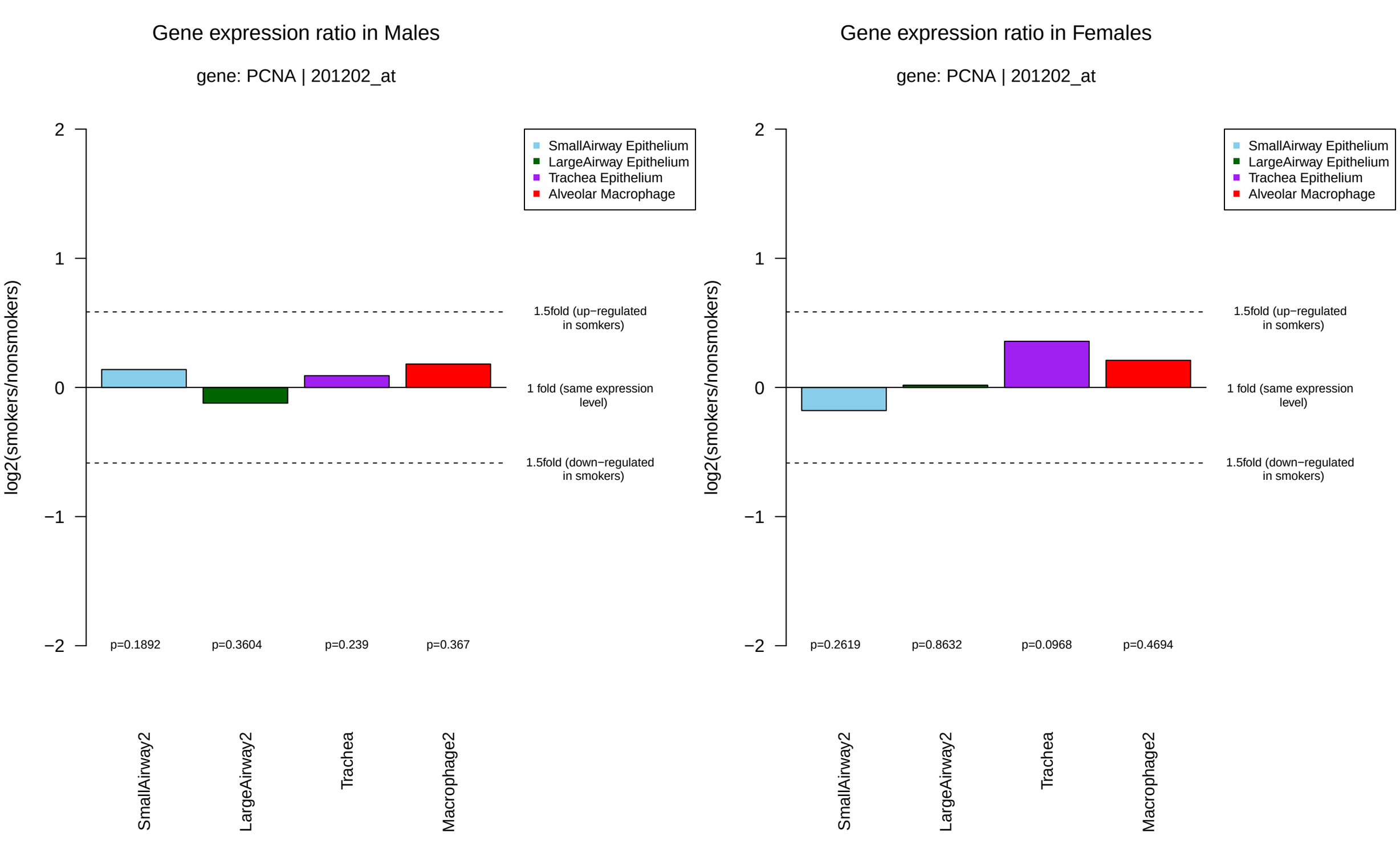
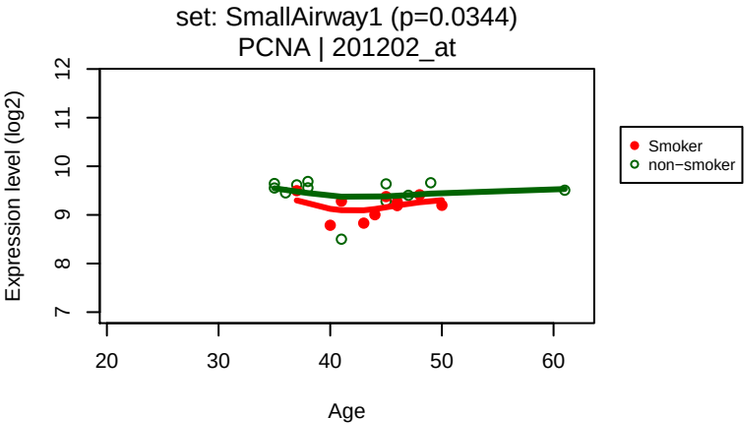
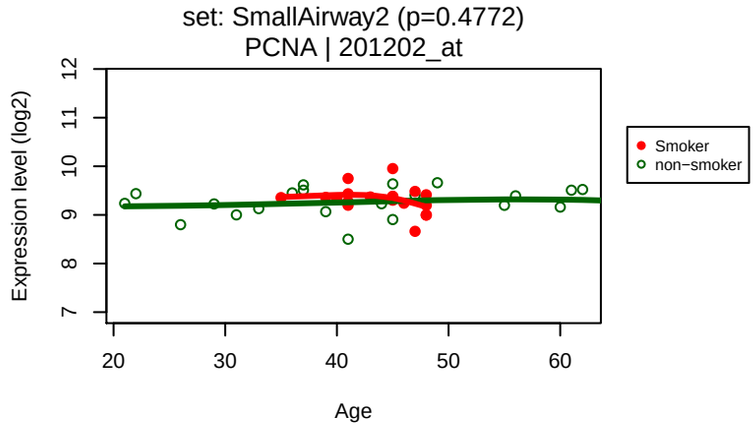
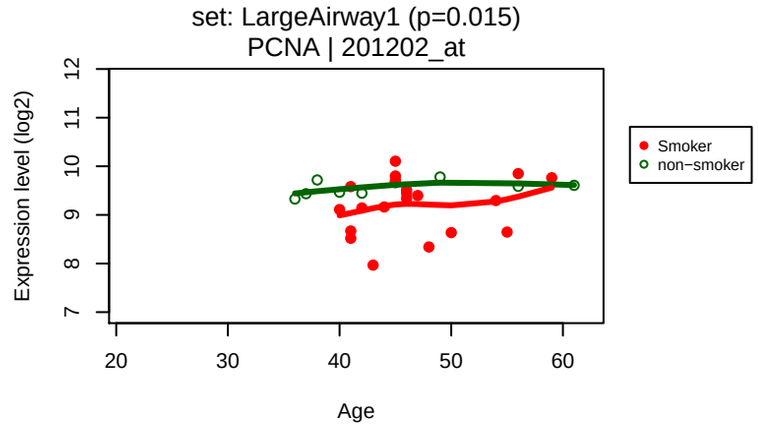

Gene expression of smokers and nonsmokers
gene: PCNA | 201202_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0344
p=0.4772
p=0.1226
p=0.015
p=0.4376
p=0.0341
p=0.8587
p=0.2109
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

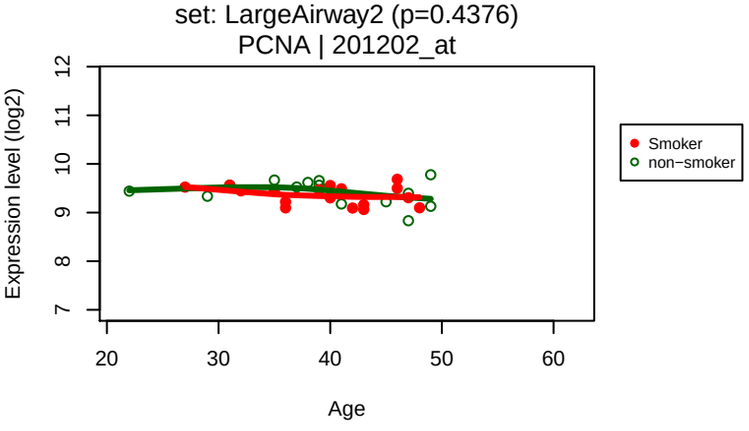

Gene expression ratio (smokers/nonsmokers)
gene: PCNA | 201202_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0344
p=0.4772
p=0.1226
p=0.015
p=0.4376
p=0.0341
p=0.8587
p=0.2109
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

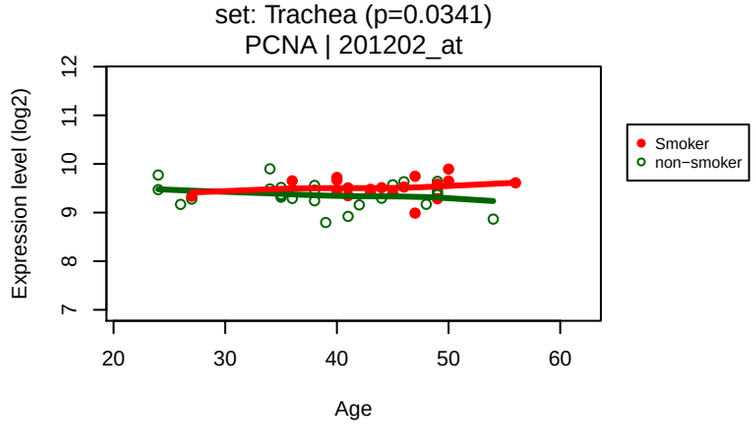

Gene expression ratio in Males
gene: PCNA | 201202_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1892
p=0.3604
p=0.239
p=0.367
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PCNA | 201202_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2619
p=0.8632
p=0.0968
p=0.4694
SmallAirway2
LargeAirway2
Trachea
Macrophage2

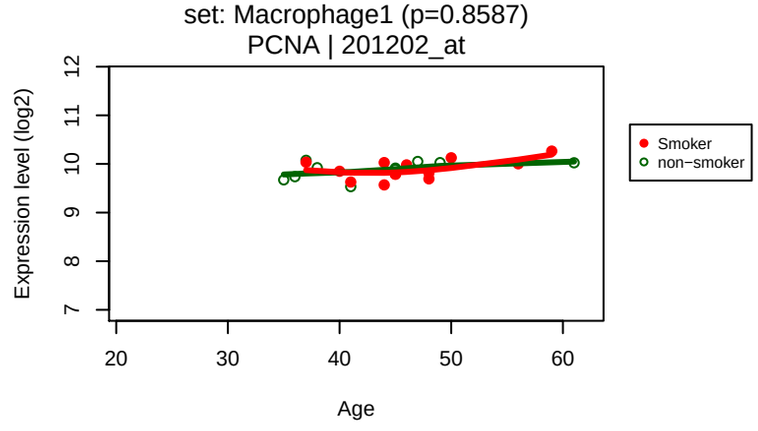

set: SmallAirway1 (p=0.0344)
PCNA | 201202_at
Expression level (log2)
Age
Smoker
non−smoker

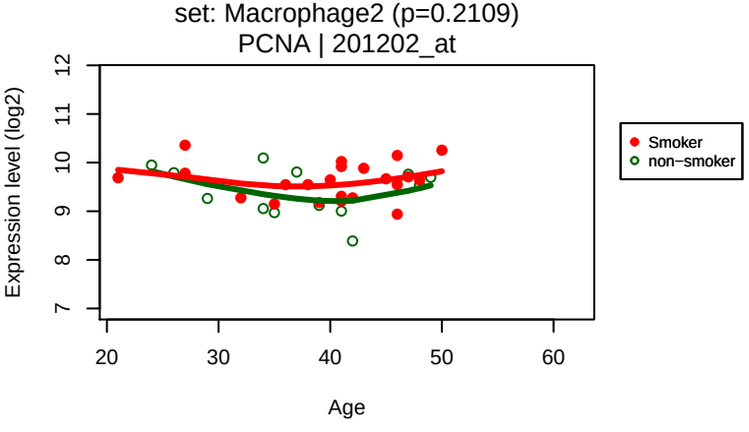

set: SmallAirway2 (p=0.4772)
PCNA | 201202_at
Expression level (log2)
Age
Smoker
non−smoker

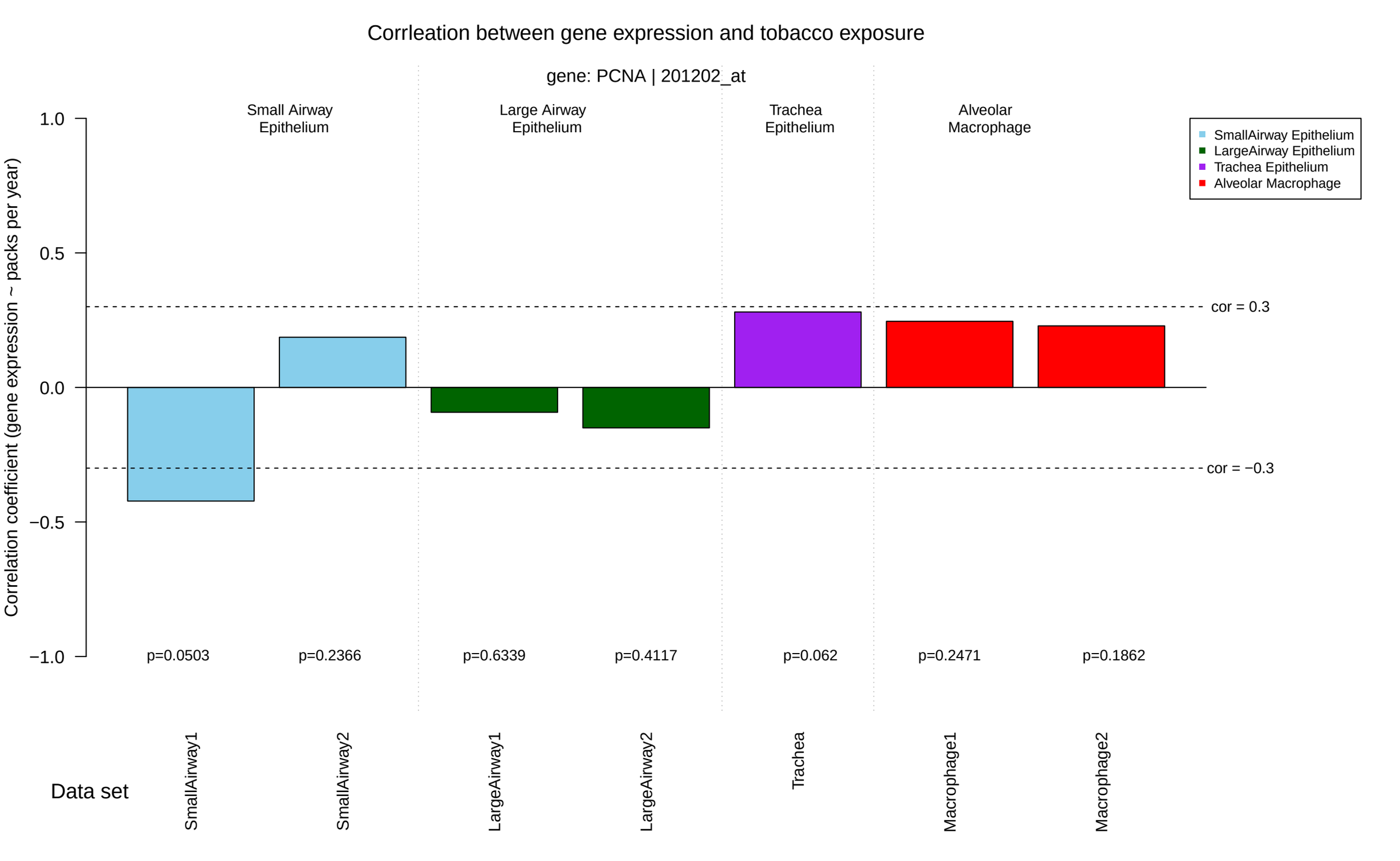

set: LargeAirway1 (p=0.015)
PCNA | 201202_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4376)
PCNA | 201202_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0341)
PCNA | 201202_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8587)
PCNA | 201202_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2109)
PCNA | 201202_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PCNA | 201202_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0503
p=0.2366
p=0.6339
p=0.4117
p=0.062
p=0.2471
p=0.1862
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2